Problem 1

Game 4

$$\int \frac{\sin^2(x)}{\cos^4(x)}\,\mathrm{d}x$$

Problem 2

Game 4

$$\int_0^1 \frac{1+\sqrt{1-x^2}}{\sqrt{1+x}+\sqrt{1-x}}\,\mathrm{d}x$$

Problem 3

Game 4

$$\int_1^\infty \frac{(-1)^{\lceil x\rceil}}{\sqrt{\lfloor x\rfloor\,\lceil x\rceil}\left(\sqrt{\lceil x\rceil}-\sqrt{\lfloor x\rfloor}\right)}\,\mathrm{d}x$$

Problem 4

Game 4

$$\int \sqrt{\frac{\ln(x+\sqrt{1+x^2})}{1+x^2}}\,\mathrm{d}x$$

Problem 5

Game 4

$$\int_0^{\frac{\pi}{4}} \frac{\cos(2x)}{-\ln(\tan(x))\cos^4(x)}\,\mathrm{d}x$$

Problem 6

Game 4

$$\int \frac{1}{1+x^2}\tan^{-1}\left(\frac{4x-4x^3}{x^4-6x^2+1}\right)\,\mathrm{d}x$$

Problem 7

Game 4

$$\lim_{n\to\infty}\int_0^{\frac{1}{2}} f_n(x)\,\mathrm{d}x$$

$$\text{where } f_n(x) = \begin{cases} 2x-2x^2, & n=1; \\ (\underbrace{f_1\circ\cdots\circ f_1}_{n\ f_1\text{'s}})(x), & n\geq 2 \end{cases}$$

Problem 8

Game 4

$$\int_0^{\infty} \frac{x\cosh(x)}{\sinh(\pi x)}\,\mathrm{d}x$$